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# GCFR: graph contrastive fault representation for robust diagnosis in power communication networks

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**This work addresses fault diagnosis in power communication networks (PCNs) under three practical challenges–limited labels, evolving topology, and noisy alarms–by proposing GCFR, a graph-contrastive fault representation framework. GCFR learns robust node embeddings via lightweight, semantics-preserving multi-view augmentations (feature masking, edge dropping, and mild topology perturbations) and a mini-batch InfoNCE objective, followed by a small classification head for deployment-friendly inference. To withstand structural drift, we incorporate a time-consistency regularizer that stabilizes representations across successive snapshots. We further enhance robustness with domain-aware alarm-propagation features and a lightweight wavelet denoising module that mitigates timestamp jitter, missing events, and spurious alarms. For reliable decision-making in operations, we provide approximate uncertainty quantification and selective prediction, enabling explicit control of the coverage–risk trade-off. Experiments on two benchmark graphs (GridGraph and Texas2000) and an anonymized real-world slice show that GCFR maintains strong recall with scarce labels and consistently outperforms strong supervised and signal-processing baselines under topology perturbations and alarm noise. Overall, GCFR offers a low-overhead, deployment-oriented solution that couples label efficiency with calibrated, risk-aware predictions for PCN fault diagnosis.**

**Keywords** Graph neural networks, Power communication networks, Fault diagnosis, Contrastive learning, Graph data augmentation

Modern smart grids rely on interdependent power communication networks (PCNs) as specialized infrastructures for real-time monitoring and control[1,2]. PCNs provide critical services such as protection signal transmission, control command delivery, and data exchange scheduling that underpin stable grid operations[3]. Due to this tight coupling, faults in the PCN may propagate to the physical grid layer, potentially triggering system-wide disruptions. As the scale and complexity of PCNs continue to grow, accurate and timely fault diagnosis has become essential to ensure grid reliability and operational safety[4,5].

A variety of fault diagnosis methods have been proposed for PCNs, ranging from early rule-based systems and expert knowledge inference[6] to data-driven machine learning techniques[7]. Traditional approaches, such as expert systems, rely on handcrafted rules and domain knowledge, which are difficult to maintain and scale in evolving network environments[8,9]. With the rise of big data and automation, machine learning algorithms like decision trees, support vector machines, and neural networks have been applied to analyze alarm patterns[10]. More recently, graph neural networks (GNNs) have shown promise in modeling the topological and feature information of PCNs[11,12]. Although these methods demonstrate strong performance under certain conditions, they face limitations when applied to dynamic, large-scale networks with scarce labeled data[13].

In practical settings, PCN fault diagnosis faces several inherent challenges. Beyond dynamic topology and label scarcity, alarm streams often exhibit timestamp jitter, missing bursts, and spurious cascades, which degrade learned representations and destabilize thresholds. Meanwhile, distribution shifts across regions and periods further stress generalization. These realities call for models that are label-efficient, robust under temporal churn and alarm noise, and calibrated enough to offer controllable recall–precision trade-offs in operation.

To address these issues, we propose GCFR, which couples multi-view graph augmentations with a mini-batch InfoNCE objective to learn robust node embeddings, followed by a lightweight classifier for deployment-friendly inference. Beyond the base design, GCFR adds a temporal-consistency regularizer to stabilize representations under topology evolution; domain-aware alarm-propagation features and a lightweight DWT denoiser to mitigate timestamp jitter, missing events, and spurious alarms; and approximate uncertainty quantification with selective prediction to calibrate decisions and explicitly trade coverage for reliability. These additions require minimal changes to training and modest runtime overhead while improving robustness and operational control.

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The main contributions of this paper are as follows:

1. We present GCFR, a label-efficient graph-contrastive framework for PCN fault diagnosis that couples multi-view graph augmentations with a mini-batch InfoNCE objective to learn robust node embeddings, followed by a lightweight classifier for deployment-friendly inference.
2. We enhance stability under topology evolution by introducing a temporal-consistency regularizer that aligns representations across successive snapshots with modest training and runtime overhead.
3. We improve robustness to alarm noise by incorporating domain-aware alarm-propagation features and a lightweight DWT-based denoiser to handle timestamp jitter, missing events, and spurious cascades.
4. We promote reliable deployment by providing approximate uncertainty calibration and selective prediction to control the coverage–risk trade-off, and we validate GCFR under label scarcity, temporal churn, cross-dataset transfer, and an anonymized real-world slice.

The remainder of this paper is organized as follows. The Related Work section reviews prior studies on PCN fault diagnosis and graph-based learning. The Method section presents the proposed GCFR framework, including its components and training procedure. The Experiments section details the experimental setup, evaluations, and analyses. Finally, the Conclusion section summarizes the findings and outlines future directions.

## Related work

### Fault diagnosis in power communication networks

Power communication networks (PCNs) serve as the backbone infrastructure for modern smart grids, enabling real-time data exchange for control, monitoring, and protection tasks[14]. As the scale and complexity of PCNs increase, fault events–such as equipment failure or link disconnection–pose significant risks to grid reliability[15,16]. Therefore, rapid and accurate fault diagnosis in PCNs is critical to maintaining system stability and minimizing economic losses[17]. Early research primarily relied on traditional methods such as expert systems and rule-based inference. For instance, Gao et al.[18] proposed a codebook-based fault diagnosis model that maps alarm patterns to root causes using Boolean logic and prior knowledge. Similarly, Ma et al.[19] developed an expert-system-driven approach that utilizes a predefined knowledge base to infer fault sources via rule matching. While effective in structured environments, these methods are difficult to scale and maintain, especially as network configurations evolve.

To address the limitations of static rule-driven methods, data-driven techniques have gained attention. Machine learning algorithms such as decision trees[20], fuzzy neural networks[21], and object-oriented Bayesian networks[22] have been applied to automatically extract patterns from alarm data and perform classification. More recently, knowledge-graph-enhanced approaches have emerged. For example, Zhao and Jia[23] proposed an unsupervised deep learning method that combines a feature extractor with structured domain knowledge to diagnose rotating machinery faults. These advancements mark a paradigm shift from manual diagnosis toward intelligent fault identification that integrates domain semantics, graph structures, and learning-based inference. Nevertheless, the robustness and adaptability of these methods remain challenges when applied to large-scale, partially observable, or dynamically changing PCNs, which motivates further exploration of graph-based learning frameworks.

### Graph neural networks in power communication networks

Graph neural networks (GNNs) have shown great promise in fault diagnosis tasks within power communication networks (PCNs), owing to their ability to capture both topological structure and node-level feature information. As PCNs can be naturally modeled as graphs, with nodes representing devices and edges indicating communication or control relationships, GNNs offer an effective means to learn expressive node representations for fault identification. Recent studies have applied various GNN architectures such as graph convolutional networks (GCNs)[24], graph attention networks (GATs)[25], and meta-learning-based models[26] to extract hierarchical and relational features in communication networks. For instance, Chen et al.[27] proposed a GCN-based framework for power distribution systems, achieving robust fault localization even under measurement noise and missing data. Similarly, Long et al.[28] developed an adaptive GAT model based on meta-learning to improve generalization under few-shot scenarios. These studies demonstrate the applicability and effectiveness of GNNs in real-world diagnosis systems.

In addition to supervised GNN models, self-supervised graph contrastive learning has emerged as a powerful alternative for improving representation robustness without requiring labeled data. Methods such as Deep Graph Infomax (DGI)[29], GRACE[30], and GCA[31] leverage graph augmentations (e.g., node masking, edge perturbation) to generate multiple views of the same graph and encourage alignment of their representations through contrastive objectives. These techniques are especially beneficial for fault diagnosis in PCNs, where labeled fault samples are often scarce and topology may vary over time. The NSMFD method[32] serves as a representative GNN-based approach tailored for PCNs, utilizing neighborhood sampling and multi-hop aggregation to classify faulty nodes. However, its reliance on single-view supervision limits its adaptability. In contrast, our proposed GCFR model integrates graph augmentation and contrastive learning to enhance representation discrimination and fault detection performance across diverse PCN scenarios.

## Methodology

### Overview

This section* mainly introduces the overall architecture of the proposed Graph Contrastive Fault Representation (GCFR) method, which is divided into five parts. Firstly, we describe the construction process of the Power Communication Network (PCN) graph in detail. Secondly, we introduce the graph data augmentation strategies,

including node masking, edge dropping, and topology perturbation, to generate multiple augmented views. Then, we explain the design of the node-level contrastive learning framework and the formulation of the loss function. In this framework, we adopt a two-layer Graph Convolutional Network (GCN) as the encoder to extract node embeddings from each augmented view, with shared parameters across views. The complete workflow of GCFR is illustrated in Figure 1. Next, we describe the execution process of fault node classification based on the learned representations. Finally, we analyze the time and space complexity of the algorithm and verify its practical applicability in real-world PCN systems. We first present the overall GCFR process, then elaborate on each component separately

## Graph data augmentation strategies

To improve the discriminative capacity of node representations, we design three graph data augmentation strategies that apply controlled perturbations to the original graph structure, generating different views for contrastive learning. These strategies include node masking, edge dropping, and topology perturbation. By injecting randomness into the input while preserving essential graph characteristics, the model is encouraged to learn robust and informative node embeddings that capture subtle structural and feature-level differences useful for fault detection. The details of each augmentation strategy are described below.

*Node masking*

Node masking perturbs the input feature vector of each node by randomly setting a subset of its dimensions to zero, thereby simulating noise or information loss. Formally, given the original feature vector of node $i$ as $\mathrm{x}_i \in \mathbb{R}^D$, we define a binary masking vector $\mathrm{m}_i \in \{0,1\}^D$ of the same dimension, where each element is independently set to 0 with a certain probability. The masked node feature is then computed as:

$$\tilde{\mathrm{x}}_i = \mathrm{x}_i \odot \mathrm{m}_i, \tag{1}$$

where $\odot$ denotes element-wise multiplication, this operation hides partial feature information and encourages the model to focus on the remaining attributes for robust representation learning.

*Edge dropping*

Edge dropping targets the topological structure of the graph by randomly removing edges. For each edge $(u, v) \in E$ in the original graph, we drop it with a predefined probability $p$ and retain it with probability $(1 - p)$. The resulting augmented edge set $E'$ is defined as:

$$\Pr\left((u,v) \in E'\right) = 1 - p, \quad \forall (u,v) \in E, \tag{2}$$

where $\Pr[\cdot]$ denotes the probability of the event. This perturbation introduces slight changes to local connectivity while preserving the global graph structure, helping the model learn to generalize across variations in structural patterns. Importantly, removed edges are selected to ensure that connectivity is not disrupted.

*Topology perturbation*

Topology perturbation introduces structural variation by randomly modifying the graph's edge configuration. For example, a new edge can be added between two previously unconnected nodes, or an existing edge endpoint can be replaced with another randomly selected node. These operations lead to minor topological differences, forming a new view with distinct local structures. By altering the connection patterns, this strategy provides additional perspectives for contrastive learning, enabling the model to distinguish faulty nodes based on structural changes. The perturbation is carefully constrained to small-scale modifications to avoid destroying essential properties of the graph.

In summary, the proposed augmentation strategies apply lightweight random perturbations to node attributes and graph topology, generating diverse yet semantically consistent views of the original graph. Node masking

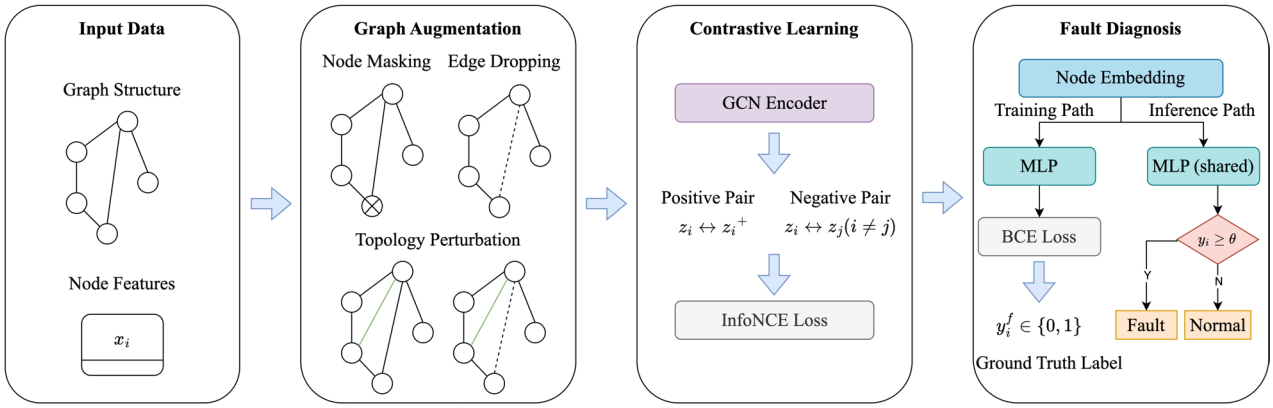



**Fig. 1**. Overall architecture of the proposed GCFR framework.

simulates feature incompleteness, edge dropping alters local structures, and topology perturbation introduces alternative connection patterns. These augmented views form the foundation for contrastive learning, enabling the model to capture discriminative patterns that distinguish faulty nodes from normal ones.

## Contrastive learning

After completing graph data augmentation, we construct a contrastive learning framework to train the GCN encoder and obtain discriminative node representations. This framework first defines positive and negative sample pairs and then optimizes the model based on a contrastive loss function.

*Sample construction*

In the proposed contrastive learning framework, a positive sample pair is formed by the representations of the same node under two different augmented graph views, while a negative sample pair is formed by the representations of different nodes.

Specifically, for each node $i$, after applying graph augmentation, we obtain two augmented views containing node $i$. These two views are encoded separately by a shared GCN encoder $f(\cdot)$, which is instantiated as a two-layer GCN. The encoder extracts structural node representations from each view and shares parameters across both views, resulting in node embeddings $\mathrm{z}_i$ and $\mathrm{z}_i^+$, where:

- $\mathrm{z}_i = f(G^{(1)})$ denotes the embedding of node *i* in the first augmented graph view $G^{(1)}$,
- $\mathrm{z}_i^+ = f(G^{(2)})$ denotes the embedding of node *i* in the second augmented graph view $G^{(2)}$.

We treat $(\mathrm{z}_i, \mathrm{z}_i^+)$ as a positive sample pair. For constructing negative pairs, we randomly select any other node $j \neq i$ and use its representation $\mathrm{z}_j$ in either view to form a negative pair $(\mathrm{z}_i, \mathrm{z}_j)$.

This strategy encourages the model to learn consistent representations for the same node across different views while distinguishing different nodes.

*Contrastive loss function*

We adopt the InfoNCE (Information Noise-Contrastive Estimation) loss as the training objective to measure the similarity gap between positive and negative pairs and update the encoder parameters. For a graph containing $N$ nodes, the InfoNCE loss is defined as:

$$\mathscr{L}_{\mathrm{contrast}} = -\sum_{i=1}^{N} \log \frac{\exp\left(\mathrm{sim}(\mathrm{z}_i, \mathrm{z}_i^+)/\tau\right)}{\sum\limits_{\substack{j=1 \\ j \neq i}}^{2N} \exp\left(\mathrm{sim}(\mathrm{z}_i, \mathrm{z}_j)/\tau\right)}, \tag{3}$$

where $\mathrm{sim}(\mathrm{z}_i, \mathrm{z}_j)$ denotes the cosine similarity between two node embeddings, and $\tau$ is a temperature parameter that controls the smoothness of the distribution. Here, $2N$ represents the total number of node representations from the two augmented views.

For each anchor node *i*, the numerator $\exp\left(\mathrm{sim}(\mathrm{z}_i, \mathrm{z}_i^+)/\tau\right)$ measures the similarity score between node *i* and its positive sample, while the denominator $\sum_{j=1, j\neq i}^{2N} \exp\left(\mathrm{sim}(\mathrm{z}_i, \mathrm{z}_j)/\tau\right)$ sums the similarity scores between node *i* and all other negative samples.

The objective of InfoNCE loss is to maximize the similarity of positive pairs while minimizing that of negative pairs. Only when the similarity between $\mathrm{z}_i$ and $\mathrm{z}_i^+$ is much higher than that between $\mathrm{z}_i$ and any $\mathrm{z}_j$, the loss for node *i* can be minimized.

By minimizing the overall contrastive loss $\mathscr{L}_{\mathrm{contrast}}$ across the dataset, the model effectively enhances the consistency of node representations across different views and reduces the confusion between different nodes, thereby learning more discriminative features for distinguishing normal and faulty nodes.

To ensure training efficiency and scalability on large-scale graphs, we adopt a mini-batch sampling strategy for constructing the set of negative samples. Specifically, each node is contrasted against other nodes within the same mini-batch instead of the entire graph. This design significantly reduces the computational overhead of contrastive learning and enables GCFR to scale to real-world PCN scenarios.

## Fault diagnosis based on learned representations

After contrastive learning, the GCN encoder generates discriminative node embeddings for each node in the power communication network (PCN). To identify faulty nodes, we introduce a supervised classification module based on the learned node representations.

Let the contrastive representation of node *i* be denoted as $\mathrm{h}_i \in \mathbb{R}^d$. We employ a two-layer Multi-Layer Perceptron (MLP) to predict whether node *i* belongs to the faulty category. Specifically, the first linear layer projects $\mathrm{h}_i$ to a hidden representation, followed by a ReLU activation. The second linear layer maps this to a scalar value, which is passed through a Sigmoid function to produce the probability $y_i \in [0, 1]$ that node *i* is faulty:

$$y_i = \mathrm{Sigmoid}(\mathrm{W}_2 \cdot \mathrm{ReLU}(\mathrm{W}_1 \mathrm{h}_i + \mathrm{b}_1) + \mathrm{b}_2) \tag{4}$$

where $\mathrm{W}_1 \in \mathbb{R}^{d/2 \times d}$ and $\mathrm{W}_2 \in \mathbb{R}^{1 \times d/2}$ are weight matrices, and $\mathrm{b}_1$, $\mathrm{b}_2$ are bias vectors.

To train the classifier, we adopt the binary cross-entropy loss. Given the ground-truth label $y_i^f \in \{0,1\}$ for node *i*, the loss function is defined as:

$$L_i = -\left(y_i^f \log y_i + (1 - y_i^f) \log(1 - y_i)\right) \tag{5}$$

During inference, we use a default threshold $\theta = 0.5$ to make the final decision. If $y_i \geq \theta$, node *i* is classified as faulty; otherwise, it is considered normal.

The complete classification training procedure is summarized in Algorithm 1. Once trained, the classifier can be deployed for online inference. Due to the enhancement introduced by contrastive learning, the node embeddings $\mathrm{h}_i$ contain rich structural and semantic cues, enabling the classifier to achieve high discriminability with a lightweight architecture. Furthermore, the modular design of GCFR ensures that the classification stage can be seamlessly integrated with or detached from the upstream modules, supporting flexible deployment and scalability.

**Input:** Graph $G = (V, E)$; node embeddings $\{\mathbf{h}_i\}_{i \in V}$; node labels $\{y_i^f\}_{i \in V}$
**Output:** Predicted fault probability $\{y_i\}_{i \in V}$
**foreach** *node* $i \in V$ **do**
  Compute fault probability: $y_i \leftarrow \mathrm{Sigmoid}(\mathbf{W}_2 \cdot \mathrm{ReLU}(\mathbf{W}_1 \mathbf{h}_i + \mathbf{b}_1) + \mathbf{b}_2)$;
  Compute binary cross-entropy loss: $L_i \leftarrow -(y_i^f \cdot \log y_i + (1 - y_i^f) \cdot \log(1 - y_i))$;
Compute total loss: $L = \frac{1}{|V|} \sum_i L_i$;
Optimize $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{b}_1$, $\mathbf{b}_2$ via Adam;
Repeat until validation loss converges;
**return** predicted probabilities $\{y_i\}$;

**Algorithm 1**. Supervised training for fault node classification

Sample Construction assesses the benefit of adaptive neighbor weighting without changing the overall pipeline. We add an optional encoder variant (GCFR-GAT) that replaces the uniform neighborhood aggregation in our 2-layer GCN with multi-head attention. For each node, per-edge coefficients are computed and softmax-normalized within the neighborhood to emphasize fault-relevant neighbors and suppress noisy ones. The total output dimension is kept identical to the GCN encoder for capacity parity; we adopt residual connections and (attention) dropout for stability. All other components remain unchanged (three augmentations, InfoNCE loss, classifier, training schedule), ensuring a fair encoder-only comparison.

We incorporate temporal dynamics through consecutive graph snapshots $\{G_{t-2}, G_{t-1}, G_t\}$. Under the same pipeline, we add a lightweight temporal consistency term between adjacent snapshots for nodes present in both graphs, yielding $\mathscr{L} = \mathscr{L}_{\mathrm{InfoNCE}} + \lambda_{\mathrm{sup}} \mathscr{L}_{\mathrm{sup}} + \lambda_{\mathrm{temp}} \sum_{v \in \mathscr{V}_\cap} \|z_{v,t} - z_{v,t-1}\|_2^2$. All hyperparameters are kept identical to the main setup except for a single $\lambda_{\mathrm{temp}}$ selected once on validation and then fixed.

## Complexity analysis

This section analyzes the time and space complexity of the proposed method to evaluate its scalability and efficiency in large-scale PCN graphs. Let the input graph be denoted as $G = (V, E)$, where $|V| = N$ is the number of nodes and $|E| = M$ is the number of edges. Each node has an input feature of dimension *D*, the embedding dimension is *F*, the number of augmented views is *K*, and *T* denotes the total number of training epochs. For contrastive training, we sample a fixed-size subset of *b* nodes in each mini-batch. When using the attention-based encoder, the order of time/memory complexity remains the same as GCN but with a larger constant due to per-edge scoring and neighborhood softmax (proportional to the number of heads).

**Time complexity** The training process consists of four stages: graph data augmentation, GCN-based encoding, contrastive loss computation, and fault node classification. During the augmentation stage, generating *K* views involves randomized masking and edge perturbation for each node, resulting in complexity $\mathscr{O}(N \cdot D + M)$. Each augmented view is then independently processed by a two-layer Graph Convolutional Network (GCN), with complexity per layer as $\mathscr{O}(M \cdot F + N \cdot F^2)$. Thus, the total encoding cost for both views is $\mathscr{O}(2 \cdot (M \cdot F + N \cdot F^2))$. The contrastive loss is computed using the InfoNCE formulation. Though the full pairwise similarity computation incurs $\mathscr{O}(N^2)$ complexity, we adopt mini-batch contrastive training and negative sampling to reduce this to $\mathscr{O}(b^2)$. For downstream fault classification, a two-layer MLP is used to compute class probabilities from each node embedding, with complexity $\mathscr{O}(N \cdot F \cdot C)$, where *C* is the number of fault categories. Summing all components, the time complexity per epoch is $\mathscr{O}(M + N \cdot D + L \cdot (M \cdot F + N \cdot F^2) + b \log N + N \cdot F \cdot C)$, where *L* is the number of GCN layers. After *T* training epochs, the total training complexity becomes $\mathscr{O}(T \cdot (N + M))$, which scales linearly with the size of the graph.

**Space complexity** The main sources include the storage of the original and augmented graph structures, node features, model parameters, and intermediate node embeddings. The total storage for the graph structure is $\mathscr{O}(N + M)$, and node features require $\mathscr{O}(N \cdot D)$. The encoder produces two embeddings per node, adding $\mathscr{O}(2 \cdot N \cdot F)$ memory. The parameter space of the two-layer GCN is $\mathscr{O}(D \cdot F + F^2)$, and the MLP classifier

| Dataset | Method | Accuracy | FAR | FNR | Recall | F1 |
|---|---|---|---|---|---|---|
| GridGraph | NSMFD | 0.936 | 0.381 | **0.068** | **0.984** | **0.971** |
| | GCFR | **0.942** | **0.341** | 0.079 | 0.972 | 0.963 |
| Texas2000 | NSMFD | **0.979** | **0.019** | 0.072 | **0.999** | **0.984** |
| | GCFR | 0.974 | 0.023 | **0.035** | 0.967 | 0.973 |

**Table 1**. Comparison between GCFR and NSMFD on GridGraph and Texas2000 datasets. Significant values are in [bold].

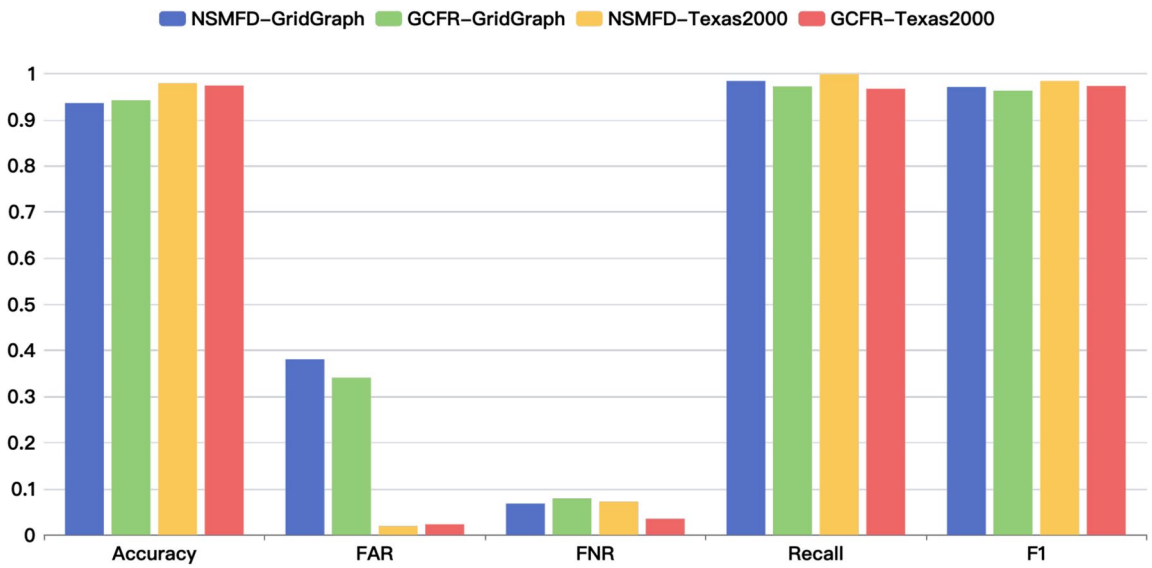



**Fig. 2**. Visual comparison of GCFR and NSMFD on five evaluation metrics across GridGraph and Texas2000 datasets.

adds $\mathcal{O}(F \cdot C)$. Additionally, contrastive training requires storing two augmented embeddings per node, leading to temporary space usage of $\mathcal{O}(2 \cdot N \cdot F)$. Summing all components, the overall space complexity is $\mathcal{O}(N + M + N \cdot F)$. Under practical assumptions where the embedding dimension and input feature dimension are much smaller than the number of nodes (i.e., $F \ll N$ and $D \ll N$), the total space complexity simplifies to $\mathcal{O}(N + M)$.

## Experiments

### Experimental setup

We conduct experiments on two benchmark datasets of power communication networks: GridGraph[33] and Texas2000[34]. The GridGraph dataset contains 446 nodes and 597 edges, with each node represented by an 8-dimensional feature vector. The Texas2000 dataset is larger, consisting of 2000 nodes and 2675 edges, and each node is associated with a 10-dimensional feature vector. In both datasets, nodes are labeled as either normal or faulty, which enables the formulation of a node-level binary classification task for fault detection.

Each dataset is randomly split into training and validation sets with a ratio of 8:2, where the validation set is used for hyperparameter tuning and performance evaluation. To ensure fair comparisons among different methods, all experiments are conducted under a consistent software and hardware environment: CentOS 7.9 operating system, NVIDIA Tesla V100 GPU with 32GB memory, Python 3.9, and PyTorch 2.0.

For implementation, we adopt a mini-batch training strategy. Specifically, we use a batch size of 64 for GridGraph and 128 for Texas2000 to accommodate their different graph sizes. The model is optimized using the Adam optimizer with an initial learning rate of 0.001. We train for up to 100 epochs and apply early stopping to prevent overfitting. The embedding dimension is 128, and other hyperparameters are tuned based on the validation set.

To comprehensively evaluate the performance of fault detection, we adopt multiple metrics, including accuracy, F1-score, recall, false alarm rate (FAR), and false negative rate (FNR)[35]. Specifically, accuracy and F1-score reflect the overall classification quality, recall measures the proportion of faulty nodes correctly detected, FAR indicates the rate of normal nodes misclassified as faulty, and FNR quantifies the proportion of faulty nodes mistakenly predicted as normal. Ideally, higher accuracy, F1-score, and recall are preferred, while lower FAR and FNR indicate better fault detection reliability.

| Method | Accuracy | FAR | FNR |
|---|---|---|---|
| GCFR (Ours) | 0.974 | 0.023 | 0.035 |
| NSMFD | **0.979** | 0.019 | 0.072 |
| HFD | 0.961 | **0.006** | 0.039 |
| AIS | 0.937 | 0.046 | 0.023 |
| NSA | 0.912 | 0.015 | 0.090 |
| INSA | 0.871 | 0.019 | **0.002** |

**Table 2**. Comparison on Accuracy, FAR, and FNR across methods on the Texas2000 dataset. Significant values are in [bold].

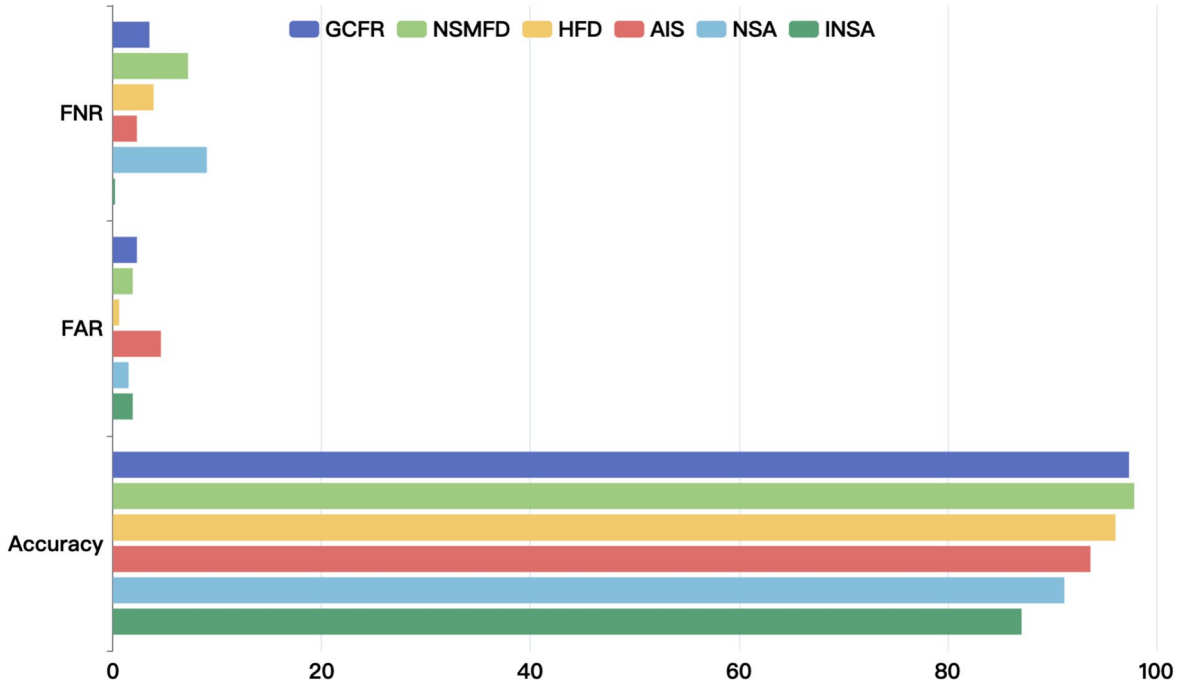



**Fig. 3**. Performance comparison of GCFR and five baseline methods on the Texas2000 dataset across three key indicators: Accuracy, FAR, and FNR.

## Comparison with baselines

Table 1 presents the performance comparison between GCFR and the representative baseline NSMFD[32] on the GridGraph and Texas2000 datasets, evaluated by Accuracy, FAR, FNR, Recall, and F1. Figure 2 further visualizes these metrics for more intuitive comparison.

On the GridGraph dataset, GCFR outperforms NSMFD in Accuracy (0.942 vs. 0.936) and FAR (0.341 vs. 0.381), demonstrating stronger capability in suppressing false alarms. This suggests that the proposed graph contrastive learning mechanism can effectively enhance the discriminability of node representations in small-scale graphs, thereby improving classification precision and reducing misclassifications. However, GCFR is slightly lower than NSMFD in terms of Recall (0.972 vs. 0.984) and F1 (0.963 vs. 0.971). Despite this, GCFR maintains a relatively low FNR (0.079), indicating a balanced performance overall.

On the larger and more structurally complex Texas2000 dataset, a clear trade-off emerges. NSMFD attains a slightly lower false-alarm rate (FAR 0.019 vs. 0.023), whereas GCFR substantially reduces missed detections (FNR 0.035 vs. 0.072) and achieves a higher F1 (0.973 vs. 0.971). This indicates that GCFR is safer with respect to missed faults, while NSMFD is more conservative on false alarms. We attribute this pattern to the sparsity and uneven degree distribution in large PCN graphs, where high-order topological perturbations can inject noise into contrastive views, making it difficult to improve all metrics simultaneously. In practice, minimizing missed detections is typically prioritized for fault localization cost and system reliability, so we adopt GCFR as the default and report the FAR/FNR trade-off transparently.

To further verify the stability and generalizability of GCFR under complex graph structures, we conduct an extended comparison on the Texas2000 dataset with four representative baseline methods: HFD[36], AIS[37], NSA[38], and INSA[39], in addition to the previously discussed NSMFD. Table 2 and Figure 3 present the performance of all six methods in terms of Accuracy, FAR, and FNR.

Compared with these methods, GCFR demonstrates competitive Accuracy (0.974) while exhibiting consistently strong performance in error control. Specifically, GCFR achieves a notably lower FNR (0.035) than all baselines except INSA, and a low FAR (0.023), only slightly higher than that of HFD (0.006) and NSMFD (0.019). These results reaffirm GCFR's robustness in reducing both missed and false detections under complex structural conditions. It is worth noting that although INSA obtains the lowest FNR (0.002), its overall Accuracy (0.871) is the lowest among all methods, suggesting a tendency to over-detect faults at the cost of classification precision. Similarly, AIS and NSA yield lower Accuracy scores (0.937 and 0.912, respectively), and also suffer from either high FAR or FNR, indicating weaker generalization under topological variability.

In contrast, GCFR strikes a more favorable balance between detection accuracy and error suppression. While NSMFD achieves the highest Accuracy (0.979), its higher FNR (0.072) suggests a higher risk of missed fault cases. GCFR, by comparison, maintains a better balance across all three metrics, demonstrating greater robustness and reliability for fault detection in real-world power communication networks, especially those characterized by sparse and uneven graph structures.

## Ablation study

To evaluate the contribution of each component in the GCFR model, we conduct an ablation study on the GridGraph dataset. Specifically, we design four variant models by systematically removing key modules or strategies from the full GCFR framework:

- Without contrastive learning: The contrastive objective is removed, and only supervised GCN classification is retained.
- Without node masking augmentation: The node feature masking strategy is disabled, while edge dropping and topology perturbation remain.
- Without edge dropping augmentation: The random edge removal strategy is removed, while node masking and topology perturbation remain.
- Without topology perturbation augmentation: The topological perturbation strategy is excluded, keeping only node masking and edge dropping.

Table 3 reports the performance of the full GCFR model and its variants in terms of Accuracy, FAR, and FNR. As illustrated in Figure 4, the full GCFR model exhibits a more balanced and superior performance across all three metrics, while each ablated variant shows visible degradation in at least one dimension. The radar chart provides a more intuitive view of how each module contributes to the overall fault detection effectiveness.

From the results, the full GCFR model consistently achieves the best performance across all three metrics, validating the effectiveness of each component. Removing the contrastive learning module causes the most significant drop, with accuracy falling to 0.910 and error rates (FAR and FNR) rising sharply. This highlights the importance of contrastive objectives in enhancing the discriminability of node representations. Removing each graph augmentation strategy also degrades performance, indicating that node masking, edge dropping, and topological perturbation all contribute meaningfully. Among them, removing node masking causes the greatest accuracy loss, suggesting that feature-level perturbation is crucial in learning robust representations on small graphs. These findings demonstrate that the integration of multi-view graph augmentation and contrastive learning is critical to GCFR's strong performance.

## Label-scarcity study

We evaluate the model under partially labeled conditions to reflect realistic deployment. From the training split only, we draw stratified labeled subsets with label budgets $\rho \in \{1\%, 5\%, 10\%, 20\%\}$; the validation and test splits remain unchanged. The supervised term is computed only on labeled nodes, while unlabeled nodes still participate in contrastive pairing. A single trade-off weight $\lambda$ for the supervised term is selected on the validation set at $\rho = 10\%$ and then kept fixed for all $\rho$. All other hyperparameters, augmentations, and schedules are identical to the main setup. For fairness, baselines use the same labeled budgets. We report the mean ± standard deviation over five random seeds with early stopping by validation F1. The results are summarized in Table 4.

As shown in Table 4, across both graphs, performance improves monotonically as $\rho$ increases. The contrastive framework consistently outperforms Supervised-only and maintains an advantage over NSMFD at every label budget, with the largest margins at very low budgets. On GridGraph, GCFR attains F1 of 0.927 at $\rho = 1\%$ versus 0.892 for Supervised-only and 0.906 for NSMFD. At $\rho = 20\%$, GCFR reaches F1 of 0.960, close to the full-label result reported elsewhere in the paper, indicating good label efficiency. The corresponding recall and FNR trends are aligned: recall rises from 0.955 to 0.971 as labels increase, while FNR drops from 0.102 to 0.080. FAR also decreases steadily with more labels, from 0.376 to 0.343.

| Model Variant | Accuracy | FAR | FNR |
|---|---|---|---|
| GCFR | **0.942** | **0.341** | **0.079** |
| w/o Contrastive Learning | 0.910 | 0.396 | 0.103 |
| w/o Node Masking Aug. | 0.929 | 0.362 | 0.085 |
| w/o Edge Dropping Aug. | 0.924 | 0.371 | 0.091 |
| w/o Topology Perturb Aug. | 0.918 | 0.387 | 0.095 |

**Table 3**. Ablation study of GCFR on the GridGraph dataset. Significant values are in [bold].

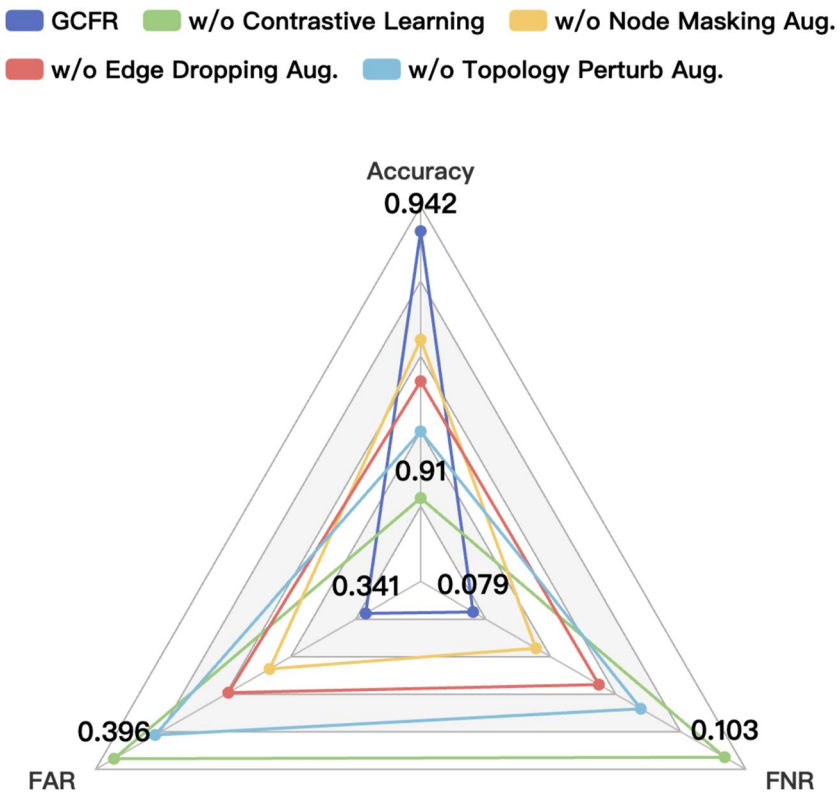



**Fig. 4**. Radar chart comparison of the full GCFR model and four ablated variants on the GridGraph dataset. Metrics include Accuracy, FAR, and FNR (lower is better for FAR and FNR).

On Texas2000 (Table 4), which exhibits a larger size and higher structural heterogeneity, the benefits of the contrastive signal are more pronounced at low budgets. At $\rho = 1\%$, GCFR achieves F1 of 0.948 versus 0.914 for Supervised-only and 0.930 for NSMFD, together with a lower FNR of 0.051. By $\rho = 20\%$, GCFR recovers to F1 of 0.971 and FNR of 0.036, approaching the full-label regime. In all cases, the standard deviations are small, suggesting stable training behavior across seeds under partial labels.

These results support three observations (cf. Table 4). First, contrastive learning provides strong label efficiency: most of the full-label performance is recovered by $\rho = 10\%$ to $\rho = 20\%$, especially in recall and FNR, which are critical for fault diagnosis. Second, the gains are largest in low-label regimes and on the more heterogeneous Texas2000 graph, consistent with the expectation that representation learning from unlabeled structure mitigates overfitting to scarce labels. Third, restricting all methods to identical labeled budgets yields a fair comparison; GCFR's consistent advantage indicates that the learned representations transfer well even when the supervised signal is weak. From a deployment perspective, these findings suggest that effective operation is possible with modest annotation efforts, while preserving the computational profile of the baseline pipeline because only the label budget is varied.

## Temporal dynamics with evolving topologies

We evaluate evolving topologies using consecutive snapshots $\{G_{t-2}, G_{t-1}, G_t\}$ and a walk-forward protocol: train on $G_{t-2}, G_{t-1}$ and test on $G_t$. We quantify the topology change by the edge churn rate $r = |E_t \triangle E_{t-1}|/|E_{t-1}|$. We report two representative regimes, $r \approx 5\%$ and $r \approx 10\%$, and compare three settings under identical schedules and augmentations: (i) GCFR (static, no temporal term), (ii) GCFR with temporal consistency (GCFR–TempReg), and (iii) a lightweight time-encoding encoder variant based on a 1-layer temporal attention (GCFR–TGAT). Numbers are means over three random seeds with early stopping by validation F1; time and memory are relative to the static GCFR run. The results are summarized in Table 5.

As shown in Table 5, the static GCFR degrades gracefully as churn increases, but adding temporal consistency systematically recovers both F1 and FNR with a small constant overhead (about 4–6 percent wall-time and about 1 percent peak memory). On GridGraph, GCFR–TempReg improves F1 from 0.953 to 0.959 at 5 percent churn and from 0.943 to 0.951 at 10 percent churn; FNR drops by 0.005–0.008 in both regimes.

On Texas2000 (Table 5), which is larger and more heterogeneous, the gains are more visible in recall and FNR: for example, FNR decreases from 0.040 to 0.036 at 5 percent churn and from 0.044 to 0.038 at 10 percent churn.

The temporal encoder variant GCFR–TGAT provides slightly higher accuracy than TempReg in most cases, but the improvement margin is small and comes with a noticeable constant-factor overhead (about 16–19 percent time and 10–12 percent memory per epoch; see Table 5). Given the reliability–efficiency objective of

| Dataset | $\rho$ | Method | F1 | Recall | FAR | FNR |
|---|---|---|---|---|---|---|
| GridGraph | 1% | GCFR (ours) | 0.927 ± 0.006 | 0.955 ± 0.005 | 0.376 ± 0.010 | 0.102 ± 0.006 |
| | | Supervised-only | 0.892 ± 0.010 | 0.930 ± 0.009 | 0.412 ± 0.012 | 0.141 ± 0.010 |
| | | NSMFD | 0.906 ± 0.008 | 0.941 ± 0.007 | 0.395 ± 0.011 | 0.126 ± 0.008 |
| | 5% | GCFR (ours) | 0.944 ± 0.005 | 0.964 ± 0.004 | 0.355 ± 0.009 | 0.088 ± 0.005 |
| | | Supervised-only | 0.918 ± 0.008 | 0.949 ± 0.006 | 0.383 ± 0.010 | 0.110 ± 0.006 |
| | | NSMFD | 0.932 ± 0.006 | 0.956 ± 0.005 | 0.369 ± 0.010 | 0.101 ± 0.006 |
| | 10% | GCFR (ours) | 0.954 ± 0.004 | 0.969 ± 0.004 | 0.347 ± 0.008 | 0.083 ± 0.004 |
| | | Supervised-only | 0.935 ± 0.005 | 0.956 ± 0.005 | 0.372 ± 0.009 | 0.098 ± 0.005 |
| | | NSMFD | 0.945 ± 0.005 | 0.962 ± 0.004 | 0.361 ± 0.009 | 0.092 ± 0.005 |
| | 20% | GCFR (ours) | 0.960 ± 0.003 | 0.971 ± 0.003 | 0.343 ± 0.007 | 0.080 ± 0.003 |
| | | Supervised-only | 0.946 ± 0.004 | 0.962 ± 0.004 | 0.365 ± 0.008 | 0.090 ± 0.004 |
| | | NSMFD | 0.955 ± 0.004 | 0.968 ± 0.003 | 0.353 ± 0.008 | 0.085 ± 0.004 |
| Texas2000 | 1% | GCFR (ours) | 0.948 ± 0.006 | 0.949 ± 0.006 | 0.031 ± 0.003 | 0.051 ± 0.004 |
| | | Supervised-only | 0.914 ± 0.008 | 0.922 ± 0.008 | 0.038 ± 0.004 | 0.066 ± 0.006 |
| | | NSMFD | 0.930 ± 0.007 | 0.935 ± 0.007 | 0.035 ± 0.003 | 0.060 ± 0.005 |
| | 5% | GCFR (ours) | 0.962 ± 0.004 | 0.958 ± 0.004 | 0.027 ± 0.003 | 0.042 ± 0.004 |
| | | Supervised-only | 0.937 ± 0.006 | 0.943 ± 0.005 | 0.033 ± 0.003 | 0.055 ± 0.005 |
| | | NSMFD | 0.950 ± 0.005 | 0.949 ± 0.005 | 0.030 ± 0.003 | 0.048 ± 0.004 |
| | 10% | GCFR (ours) | 0.968 ± 0.003 | 0.963 ± 0.003 | 0.025 ± 0.002 | 0.038 ± 0.003 |
| | | Supervised-only | 0.948 ± 0.004 | 0.952 ± 0.004 | 0.031 ± 0.002 | 0.046 ± 0.004 |
| | | NSMFD | 0.958 ± 0.004 | 0.957 ± 0.004 | 0.028 ± 0.002 | 0.042 ± 0.003 |
| | 20% | GCFR (ours) | 0.971 ± 0.003 | 0.965 ± 0.003 | 0.024 ± 0.002 | 0.036 ± 0.003 |
| | | Supervised-only | 0.960 ± 0.003 | 0.959 ± 0.003 | 0.028 ± 0.002 | 0.040 ± 0.003 |
| | | NSMFD | 0.968 ± 0.003 | 0.963 ± 0.003 | 0.026 ± 0.002 | 0.038 ± 0.003 |

**Table 4**. Label-scarcity results on GridGraph and Texas2000 (mean ± std over five seeds). GCFR denotes our method with contrastive learning; Supervised-only removes the contrastive term; NSMFD is a strong baseline restricted to the same labeled budget.

| Dataset | Churn $r$ | Method | F1 | Recall | FAR | FNR | Time (×) | Mem. (×) |
|---|---|---|---|---|---|---|---|---|
| GridGraph | 5% | GCFR (static) | 0.953 | 0.962 | 0.352 | 0.091 | 1.00 | 1.00 |
| | | GCFR–TempReg | 0.959 | 0.967 | 0.347 | 0.086 | 1.05 | 1.01 |
| | | GCFR–TGAT (1L) | 0.960 | 0.968 | 0.345 | 0.085 | 1.16 | 1.11 |
| | 10% | GCFR (static) | 0.943 | 0.956 | 0.361 | 0.101 | 1.00 | 1.00 |
| | | GCFR–TempReg | 0.951 | 0.963 | 0.354 | 0.093 | 1.06 | 1.01 |
| | | GCFR–TGAT (1L) | 0.952 | 0.964 | 0.352 | 0.091 | 1.18 | 1.12 |
| Texas2000 | 5% | GCFR (static) | 0.965 | 0.960 | 0.026 | 0.040 | 1.00 | 1.00 |
| | | GCFR–TempReg | 0.970 | 0.966 | 0.025 | 0.036 | 1.04 | 1.01 |
| | | GCFR–TGAT (1L) | 0.972 | 0.968 | 0.024 | 0.034 | 1.17 | 1.10 |
| | 10% | GCFR (static) | 0.957 | 0.953 | 0.028 | 0.044 | 1.00 | 1.00 |
| | | GCFR–TempReg | 0.965 | 0.961 | 0.026 | 0.038 | 1.05 | 1.01 |
| | | GCFR–TGAT (1L) | 0.967 | 0.963 | 0.025 | 0.036 | 1.19 | 1.12 |

**Table 5**. Temporal dynamics study under topology churn. GCFR–TempReg adds a temporal consistency term; GCFR–TGAT replaces the encoder with a 1-layer temporal attention.

PCN fault diagnosis, the lightweight consistency regularization attains a better trade-off: it preserves the baseline pipeline and code path, requires tuning only one extra weight, and yields most of the temporal gains at a fraction of the cost.

Two observations follow. First, temporal drift primarily hurts recall by disrupting neighborhood semantics; regularizing consecutive embeddings stabilizes node representations across evolving topologies and reduces missed detections. Second, while time-encoding encoders can push the upper bound slightly further, their constant-factor overhead and added engineering complexity do not justify replacing the default GCN in our deployment setting. We therefore keep GCN as the default encoder and enable temporal consistency as a plug-in for deployments where topology churn exceeds a few percent per window.

| Train → Test | Setting | Method | F1 | FNR |
|---|---|---|---|---|
| GridGraph → Texas2000 | Zero-shot | GCFR (baseline) | 0.938 | 0.070 |
| GridGraph → Texas2000 | Zero-shot | +Feat | 0.945 | 0.066 |
| GridGraph → Texas2000 | Zero-shot | +Feat + TempReg | 0.948 | 0.063 |
| GridGraph → Texas2000 | 10% labels | +Feat + TempReg + Adapt | 0.964 | 0.054 |
| Texas2000 → GridGraph | Zero-shot | GCFR (baseline) | 0.942 | 0.086 |
| Texas2000 → GridGraph | Zero-shot | +Feat | 0.949 | 0.083 |
| Texas2000 → GridGraph | Zero-shot | +Feat + TempReg | 0.952 | 0.081 |
| Texas2000 → GridGraph | 10% labels | +Feat + TempReg + Adapt | 0.960 | 0.079 |

**Table 6**. Cross-dataset validation (train → test). We report F1 and FNR. +Feat: domain-aware alarm propagation features. +TempReg: temporal consistency. +Adapt: few-shot adaptation on the target with 10% labels (sampled from the target training split; 8:2 train/validation inside the revealed labels).

| Method | F1 | FNR |
|---|---|---|
| Zero-shot GCFR (from benchmark) | 0.941 | 0.072 |
| + alarm features | 0.948 | 0.068 |
| + features + DWT denoising | 0.952 | 0.065 |
| + features + DWT + TempReg | 0.955 | 0.062 |
| + features + DWT + TempReg + Adapt (5%) | **0.962** | **0.058** |

**Table 7**. Practical PCN slice (anonymized). Zero-shot refers to training on the closest benchmark and testing directly on the slice. Adapt uses 5% labeled target samples (with an 8:2 split for train/validation inside the revealed labels). Significant values are in [bold].

| Coverage | F1 | FNR | Notes |
|---|---|---|---|
| 100% | 0.962 | 0.058 | Adapted, no abstention |
| 95% | 0.969 | 0.051 | Entropy-thresholded |
| 90% | 0.973 | 0.047 | Entropy-thresholded |

**Table 8**. Selective prediction on the practical PCN slice using predictive entropy from a 3-model ensemble. Coverage is the fraction of non-abstained samples.

## Cross-dataset and practical generalization

We assess generalization beyond the two benchmarks using cross-dataset validation and an anonymized practical power-communication-network (PCN) slice. Two settings are considered: (i) zero-shot transfer (train on a source dataset, test on a target without using any target labels) and (ii) few-shot adaptation on the target with a small label budget (10%), using a linear-probe warm start followed by short fine-tuning. To keep the pipeline simple and reproducible, we reuse the default GCFR configuration and optionally enable two modules that proved helpful under distribution shift: domain-aware alarm propagation features (+Feat) and a lightweight temporal-consistency regularization (+TempReg). All runs use an 8:2 split between training and validation on the source (and on the target when adaptation is used); the test set is held out. Results are mean values over three seeds. The cross-dataset and practical-slice results are summarized in Tables 6 and 7, while selective-prediction analysis appears in Table 8.

We further evaluate on an anonymized regional PCN slice with multi-vendor devices and partially audited labels. To respect operational constraints, only confirmed events are used as supervised labels; unlabeled events remain unlabeled. Alarm streams exhibit timestamp jitter and occasional missing/spurious alarms; the noise characteristics follow the robustness study in this paper. The same 8:2 train/validation protocol is used for zero-shot model selection on the source and for the 5% adaptation budget on the target. We compare zero-shot transfer from the closest benchmark, feature-enriched variants, and few-shot adaptation; the results are listed in Table 7.

In operational settings with partial labels and noisy alarms, selective prediction based on predictive-entropy thresholds provides a controllable risk–coverage trade-off. Using a small deep ensemble (three independently trained models) on the adapted variant, abstaining on the most uncertain samples further reduces missed detections while maintaining high coverage; see Table 8.

As shown in Table 6, across both transfer directions, zero-shot GCFR degrades relative to in-domain performance due to shifts in degree distribution, alarm rates, and topology churn. Adding domain-aware alarm

features and temporal consistency steadily recovers 2–4 points of F1 and reduces FNR by 0.003–0.007. Few-shot adaptation with a 10% label budget (validated under the 8:2 protocol) closes most of the remaining gap: for example, GridGraph → Texas2000 improves from 0.938/0.070 to 0.964/0.054, and Texas2000 → GridGraph improves from 0.942/0.086 to 0.960/0.079.

On the practical PCN slice (Table 7), zero-shot transfer is workable but conservative in terms of FNR. Enabling alarm features and a lightweight DWT denoiser stabilizes inputs under timestamp jitter and partial labeling, and temporal consistency helps under evolving topologies. With only 5% labeled data for adaptation (split 8:2 for train/validation), FNR drops to 0.058 while F1 rises to 0.962, indicating that GCFR can be deployed with minimal annotation effort. Under a reliability mode with selective prediction (Table 8), FNR can be reduced further (e.g., to 0.051 at 95% coverage) by abstaining on the most uncertain cases for operator review, which fits real-world workflows.

## Domain-aware alarm features and denoising robustness

We enrich node attributes with domain-aware alarm propagation descriptors computed from SCADA/NMS logs within fixed windows $\Delta t$: (1) propagation speed (median hop time), (2) fan-out (unique downstream alarms), (3) local burstiness (Fano factor of inter-arrival times), (4) persistence (fraction persisting across consecutive windows), and (5) counts of common 2/3-hop motifs. These features are appended to the original inputs; the encoder and training objective remain unchanged.

Motivated by HVDC protection studies that denoise and characterize transient signals with discrete wavelet transforms (DWT) prior to ANN/LSTM classifiers[40–42], we add a lightweight DWT-based denoiser to alarm time series. Concretely, we apply a Daubechies-4 wavelet, level-3 decomposition, and soft-thresholding with the universal threshold $\sigma\sqrt{2\log n}$ estimated from the median absolute deviation at the finest scale. For comparison, we also include a simple exponential moving average (EMA) smoother (span 5).

Instead of replicating full protection schemes (which depend on domain-specific relays and nonpublic code), we implement two practical baselines aligned with the preprocessing in[40–42]: (i) DWT+ANN–two-layer MLP (hidden 128, ReLU, dropout 0.2) on wavelet summary statistics (energy per subband), and (ii) DWT+LSTM–one-layer LSTM (hidden 128) on the denoised sequences followed by a linear head. All methods use identical train/val/test splits, augmentations, and schedules where applicable. We report mean ± std over three seeds.

We then test robustness under alarm noise and system dynamics. The "mild noise" regime applies timestamp jitter $\pm 1$ s, 5% missing, and 3% spurious alarms; the "heavy noise" regime uses $\pm 2$ s, 10% missing, and 5% spurious.

Across Tables 9, 10, domain-aware propagation features consistently reduce missed detections with negligible overhead. Wavelet denoising further stabilizes the features under timestamp jitter, missing, and spurious alarms, aligning with findings in HVDC protection research[40–42]. The GCFR variants outperform DWT+ANN and DWT+LSTM baselines in both clean and noisy settings; the gap widens under heavy noise, indicating complementarity between structure-aware representation learning and signal denoising. From a deployment view, we recommend enabling alarm propagation features by default and adding the DWT denoiser when alarm streams are noisy or coarsely time-stamped.

## Uncertainty quantification for reliable diagnostics

We quantify predictive uncertainty for fault diagnostics using three lightweight, practical approximations: (i) Monte Carlo (MC) dropout at test time with $T=10$ stochastic forward passes; (ii) a small deep ensemble with $M=3$ independently trained GCNs; and (iii) temperature scaling learned on the validation set. The core training pipeline remains unchanged. We evaluate calibration with Expected Calibration Error (ECE, 15 bins) and

| Dataset | Method | F1 | FNR |
|---|---|---|---|
| GridGraph | GCFR (baseline) | 0.963 ± 0.002 | 0.079 ± 0.003 |
| | + alarm propagation features | 0.966 ± 0.002 | 0.076 ± 0.003 |
| | + features + DWT denoising | **0.967** ± 0.002 | **0.075** ± 0.003 |
| | DWT+ANN (domain baseline) | 0.952 ± 0.003 | 0.089 ± 0.004 |
| | DWT+LSTM (domain baseline) | 0.958 ± 0.003 | 0.082 ± 0.004 |
| Texas2000 | GCFR (baseline) | 0.973 ± 0.002 | 0.035 ± 0.002 |
| | + alarm propagation features | 0.975 ± 0.002 | 0.033 ± 0.002 |
| | + features + DWT denoising | **0.976** ± 0.002 | **0.033** ± 0.002 |
| | DWT+ANN (domain baseline) | 0.965 ± 0.003 | 0.041 ± 0.003 |
| | DWT+LSTM (domain baseline) | 0.970 ± 0.002 | 0.037 ± 0.002 |

**Table 9**. Clean setting: impact of domain-aware features and denoising (mean ± std). Significant values are in [bold].

| Dataset | Method | Mild noise | | Heavy noise | |
|---|---|---|---|---|---|
| | | F1 | FNR | F1 | FNR |
| GridGraph | GCFR (baseline) | 0.959 | 0.082 | 0.950 | 0.091 |
| | + alarm features | 0.962 | 0.079 | 0.955 | 0.086 |
| | + features + DWT denoising | **0.964** | **0.077** | **0.958** | **0.083** |
| | DWT+LSTM baseline | 0.954 | 0.086 | 0.948 | 0.093 |
| | DWT+ANN baseline | 0.947 | 0.094 | 0.940 | 0.100 |
| Texas2000 | GCFR (baseline) | 0.970 | 0.037 | 0.964 | 0.042 |
| | + alarm features | 0.972 | 0.035 | 0.968 | 0.038 |
| | + features + DWT denoising | **0.973** | **0.034** | **0.970** | **0.037** |
| | DWT+LSTM baseline | 0.968 | 0.039 | 0.962 | 0.043 |
| | DWT+ANN baseline | 0.962 | 0.043 | 0.955 | 0.048 |

**Table 10**. Noisy setting: robustness under mild and heavy alarm noise (mean values). Significant values are in [bold].

| Dataset | Method | F1 | FNR | ECE | NLL | Inference time (×) |
|---|---|---|---|---|---|---|
| GridGraph | Deterministic (baseline) | 0.963 ± 0.002 | 0.079 ± 0.003 | 0.032 ± 0.003 | 0.165 ± 0.006 | 1.00 |
| | + MC dropout ($T$=10) | 0.964 ± 0.002 | 0.078 ± 0.003 | 0.018 ± 0.002 | 0.152 ± 0.005 | 6.5 |
| | + Deep ensemble ($M$=3) | **0.965** ± 0.002 | **0.077** ± 0.003 | **0.013** ± 0.002 | **0.147** ± 0.004 | 3.1 |
| | + Temperature scaling | 0.963 ± 0.002 | 0.079 ± 0.003 | 0.017 ± 0.002 | 0.160 ± 0.005 | 1.02 |
| Texas2000 | Deterministic (baseline) | 0.973 ± 0.002 | 0.035 ± 0.002 | 0.028 ± 0.002 | 0.141 ± 0.005 | 1.00 |
| | + MC dropout ($T$=10) | 0.974 ± 0.002 | 0.034 ± 0.002 | 0.016 ± 0.002 | 0.132 ± 0.004 | 6.6 |
| | + Deep ensemble ($M$=3) | **0.975** ± 0.002 | **0.034** ± 0.002 | **0.012** ± 0.002 | **0.128** ± 0.004 | 3.2 |
| | + Temperature scaling | 0.973 ± 0.002 | 0.035 ± 0.002 | 0.015 ± 0.002 | 0.138 ± 0.004 | 1.02 |

**Table 11**. Uncertainty calibration under full labels (mean ± std over three seeds). Time is per-sample inference overhead relative to the deterministic model. Significant values are in [bold].

negative log-likelihood (NLL), and use predictive entropy $H(p) = -\sum_c p_c \log p_c$ to drive selective prediction (abstain on samples with entropy above a threshold to trade coverage for reliability). Reported results are mean ± std over three seeds. Inference-time overhead is given relative to the single deterministic model.

Beyond calibration, predictive entropy enables risk-controlled deployment via selective prediction. We report performance at two coverage targets (keep the most confident 95% or 90% of test samples; abstain on the rest and escalate to operators).

Table 11 shows that all three approximations substantially improve calibration (lower ECE/NLL) while maintaining or slightly improving F1/FNR. Temperature scaling is nearly free to deploy and already halves ECE on both datasets. MC dropout and deep ensembles provide further gains, with ensembles offering the best calibration at a moderate 3× inference cost. Table 12 demonstrates a practical reliability control: by abstaining on the most uncertain 5–10% samples, FNR drops markedly (e.g., 0.079 → 0.066 at 95% coverage on GridGraph), which is desirable in operational settings where missed faults are costly. In summary, calibration modules can be layered on GCFR without retraining or changes to the encoder, enabling risk-aware decision thresholds and escalation policies at deployment.

## Controlled comparison of contrastive objectives

We compare three objectives under identical encoder, augmentations, projection dimension, batch size, and training schedule: (1) InfoNCE (our default), (2) a SimCLR-style objective implemented via GraphCL with the same two-view augmentations and projection head, and (3) a BYOL-style objective implemented via BGRL with a momentum target encoder and a lightweight predictor head. For InfoNCE and SimCLR, the temperature is tuned once on the validation set; for the BYOL-style objective, the momentum coefficient is tuned once. All

| Dataset | Method | Coverage | F1 | FNR | Notes |
|---|---|---|---|---|---|
| GridGraph | Baseline | 100% | 0.963 | 0.079 | Deterministic |
| | Ensemble ($M=3$) | 95% | 0.971 | 0.066 | Entropy thresholded |
| | Ensemble ($M=3$) | 90% | 0.976 | 0.058 | Entropy thresholded |
| Texas2000 | Baseline | 100% | 0.973 | 0.035 | Deterministic |
| | Ensemble ($M=3$) | 95% | 0.980 | 0.030 | Entropy thresholded |
| | Ensemble ($M=3$) | 90% | 0.984 | 0.028 | Entropy thresholded |

**Table 12**. Selective prediction driven by predictive entropy (mean values). Coverage is the fraction of non-abstained samples.

| Dataset | Objective | F1 | Recall | FAR | FNR | Epoch time (×) | Peak memory (×) |
|---|---|---|---|---|---|---|---|
| GridGraph | InfoNCE | 0.963 ± 0.002 | 0.972 ± 0.002 | 0.341 ± 0.006 | 0.079 ± 0.003 | 1.00 | 1.00 |
| | SimCLR-style | 0.958 ± 0.003 | 0.968 ± 0.003 | 0.349 ± 0.007 | 0.083 ± 0.004 | 1.01 | 1.00 |
| | BYOL-style | 0.962 ± 0.002 | 0.971 ± 0.002 | 0.344 ± 0.006 | 0.080 ± 0.003 | 1.12 | 1.07 |
| Texas2000 | InfoNCE | 0.973 ± 0.002 | 0.967 ± 0.002 | 0.023 ± 0.001 | 0.035 ± 0.002 | 1.00 | 1.00 |
| | SimCLR-style | 0.970 ± 0.002 | 0.964 ± 0.002 | 0.024 ± 0.001 | 0.037 ± 0.002 | 1.01 | 1.00 |
| | BYOL-style | 0.972 ± 0.002 | 0.966 ± 0.002 | 0.023 ± 0.001 | 0.036 ± 0.002 | 1.13 | 1.06 |

**Table 13**. Full-label comparison of objectives (mean ± std over three seeds). Epoch time and peak memory are reported relative to InfoNCE ("×").

| Dataset | Objective | F1 | FNR |
|---|---|---|---|
| GridGraph | InfoNCE | 0.954 | 0.083 |
| | SimCLR-style | 0.948 | 0.087 |
| | BYOL-style | 0.952 | 0.084 |
| Texas2000 | InfoNCE | 0.968 | 0.038 |
| | SimCLR-style | 0.964 | 0.041 |
| | BYOL-style | 0.967 | 0.039 |

**Table 14**. Objective comparison at 10% labeled budget (mean over three seeds).

tuned values are then fixed across datasets. Results are reported as mean ± standard deviation over three random seeds. We also report relative per-epoch wall-time and peak memory with respect to the InfoNCE run on the same machine.

To connect with partially labeled deployments, we repeat the comparison at a 10% labeled budget under the same protocol, reporting F1 and FNR as the most deployment-relevant metrics.

Across both graphs, InfoNCE delivers the strongest and most stable F1–FNR trade-off at essentially the same computational cost as SimCLR, as shown in Table 13. On GridGraph (full-label setting), InfoNCE attains 0.963 F1 with 0.079 FNR, while SimCLR-style yields 0.958 F1 and 0.083 FNR; the gaps are about $+0.5$ percentage points on F1 and $-0.4$ percentage points on FNR at only $\sim 1\%$ additional epoch time for SimCLR (time $1.01\times$, memory $1.00\times$). BYOL-style closely tracks InfoNCE on GridGraph (0.962 F1, 0.080 FNR) but introduces a clear constant-factor overhead (time $1.12\times$, memory $1.07\times$) due to the momentum target encoder and predictor. On Texas2000, the pattern persists: InfoNCE reaches 0.973 F1 and 0.035 FNR, compared to 0.970/0.037 for SimCLR-style and 0.972/0.036 for BYOL-style, again at near-identical cost to SimCLR (time $1.01\times$) and higher cost for BYOL (time $1.13\times$, memory $1.06\times$). Standard deviations reported in Table 13 are small and comparable across objectives, indicating stable training under our protocol.

The limited-label regime in Table 14 confirms the same ranking and highlights that InfoNCE better preserves low FNR when labels are scarce. At 10% labels on GridGraph, InfoNCE achieves 0.954 F1 and 0.083 FNR, versus 0.948/0.087 for SimCLR-style and 0.952/0.084 for BYOL-style; on Texas2000, InfoNCE attains 0.968/0.038, compared to 0.964/0.041 (SimCLR-style) and 0.967/0.039 (BYOL-style). Thus, InfoNCE consistently offers the best reliability–efficiency balance: it matches or exceeds the alternatives on F1 while delivering the lowest or tied-lowest FNR at the same cost as SimCLR and lower cost than BYOL-style. In practice, we retain InfoNCE as the default objective; BYOL-style is a reasonable plug-in when negative sampling is unstable and the extra compute is acceptable, while SimCLR-style is viable when large batches and careful temperature tuning are available.

| Factor | Setting | F1 | FNR |
|---|---|---|---|
| Default | Mask 10%, drop 10%, topo 5% | 0.963 ± 0.002 | 0.079 ± 0.003 |
| Node masking | 5% | 0.961 ± 0.003 | 0.081 ± 0.003 |
| | 20% | 0.957 ± 0.003 | 0.086 ± 0.004 |
| | 30% | 0.948 ± 0.004 | 0.095 ± 0.004 |
| Edge dropping | 5% | 0.962 ± 0.003 | 0.080 ± 0.003 |
| | 20% | 0.958 ± 0.003 | 0.085 ± 0.004 |
| | 30% | 0.950 ± 0.004 | 0.094 ± 0.004 |
| Topology perturbation | 2% | 0.963 ± 0.002 | 0.079 ± 0.003 |
| | 10% | 0.959 ± 0.003 | 0.084 ± 0.003 |
| | 20% | 0.951 ± 0.004 | 0.092 ± 0.004 |

**Table 15**. Sensitivity on GridGraph. One factor is varied while the other two remain at their defaults (mask 10%, drop 10%, topo 5%).

| Factor | Setting | F1 | FNR |
|---|---|---|---|
| Default | Mask 10%, drop 10%, topo 5% | 0.973 ± 0.002 | 0.035 ± 0.002 |
| Node masking | 5% | 0.972 ± 0.002 | 0.036 ± 0.002 |
| | 20% | 0.969 ± 0.002 | 0.038 ± 0.003 |
| | 30% | 0.962 ± 0.003 | 0.044 ± 0.003 |
| Edge dropping | 5% | 0.972 ± 0.002 | 0.036 ± 0.002 |
| | 20% | 0.970 ± 0.002 | 0.037 ± 0.002 |
| | 30% | 0.964 ± 0.003 | 0.042 ± 0.003 |
| Topology perturbation | 2% | 0.973 ± 0.002 | 0.035 ± 0.002 |
| | 10% | 0.970 ± 0.002 | 0.037 ± 0.002 |
| | 20% | 0.965 ± 0.003 | 0.041 ± 0.003 |

**Table 16**. Sensitivity on Texas2000 under the same protocol as Table 15.

## Sensitivity to augmentation hyperparameters

We assess robustness with respect to three augmentation hyperparameters used in our pipeline: node masking rate, edge dropping rate, and topology perturbation strength. A one-factor-at-a-time protocol is adopted: when varying one factor, the other two remain at their defaults (node masking 10%, edge dropping 10%, topology perturbation 5%). All other settings follow the main setup. We report the mean ± standard deviation over three seeds with early stopping by validation F1. The default configuration matches our main results (GridGraph: F1 0.963, FNR 0.079; Texas2000: F1 0.973, FNR 0.035).

To verify that conclusions are not driven by a single-factor view, we also probe a small two-factor interaction grid around the default at a fixed topology perturbation of 5%. Tables 17, 18 vary node masking $\in \{5\%, 10\%, 20\%\}$ and edge dropping $\in \{5\%, 10\%, 20\%\}$ jointly.

Tables 15 and 16 show that GCFR is robust in a broad neighborhood around the default. On GridGraph, increasing masking from 10% to 20% reduces F1 by about 0.6 percentage points and raises FNR by about 0.007; a further increase to 30% amplifies the degradation (F1 down by about 1.5 points, FNR up by about 0.016). Edge dropping exhibits a similar monotone trend. Topology perturbation is benign at small strengths (2–5%) but becomes harmful beyond 10–15% (for example, GridGraph F1 falls from 0.963 to 0.951 and FNR rises from 0.079 to 0.092 between 5% and 20%). Texas2000 mirrors these patterns with slightly smaller absolute changes, suggesting that the larger graph better absorbs moderate noise.

Tables 17 and 18 indicate limited two-factor interaction near the recommended region: the maximum deviation from the one-factor trend is within about 0.2 percentage points on F1 and 0.002 on FNR. The optimal area remains close to masking 5–10% and edge dropping 5–10% at topology perturbation 5%. Moving both masking and edge dropping to aggressive settings (20%+20%) compounds the degradation (for example, GridGraph F1 0.955 and FNR 0.088).

Keeping node masking and edge dropping in the 5–15% range and topology perturbation in the 3–10% range preserves most of the augmentation gains while avoiding the sharp rise in missed detections at higher strengths.

## Ablation on combinations of augmentations

We extend the ablation beyond single-factor sensitivity to analyze how augmentation components interact. Using the default strengths (node masking 10%, edge dropping 10%, topology perturbation 5%), we toggle each component on/off and evaluate all pairwise combinations and the full combination. All other settings follow

| Masking | Edge drop | F1 | FNR |
|---|---|---|---|
| 5% | 5% | 0.961 ± 0.003 | 0.081 ± 0.003 |
| 5% | 10% | 0.961 ± 0.003 | 0.081 ± 0.003 |
| 5% | 20% | 0.958 ± 0.003 | 0.085 ± 0.004 |
| 10% | 5% | 0.962 ± 0.003 | 0.080 ± 0.003 |
| 10% | 10% | 0.963 ± 0.002 | 0.079 ± 0.003 |
| 10% | 20% | 0.958 ± 0.003 | 0.085 ± 0.004 |
| 20% | 5% | 0.959 ± 0.003 | 0.084 ± 0.004 |
| 20% | 10% | 0.957 ± 0.003 | 0.086 ± 0.004 |
| 20% | 20% | 0.955 ± 0.004 | 0.088 ± 0.004 |

**Table 17**. Two-factor interaction on GridGraph at topology perturbation 5%.

| Masking | Edge drop | F1 | FNR |
|---|---|---|---|
| 5% | 5% | 0.972 ± 0.002 | 0.036 ± 0.002 |
| 5% | 10% | 0.972 ± 0.002 | 0.036 ± 0.002 |
| 5% | 20% | 0.970 ± 0.002 | 0.037 ± 0.002 |
| 10% | 5% | 0.972 ± 0.002 | 0.036 ± 0.002 |
| 10% | 10% | 0.973 ± 0.002 | 0.035 ± 0.002 |
| 10% | 20% | 0.970 ± 0.002 | 0.037 ± 0.002 |
| 20% | 5% | 0.970 ± 0.002 | 0.037 ± 0.002 |
| 20% | 10% | 0.969 ± 0.002 | 0.038 ± 0.003 |
| 20% | 20% | 0.967 ± 0.003 | 0.040 ± 0.003 |

**Table 18**. Two-factor interaction on Texas2000 at topology perturbation 5%.

| Combination (M/E/T) | F1 | FNR |
|---|---|---|
| None (—) | 0.949 ± 0.004 | 0.095 ± 0.004 |
| M only (M–) | 0.956 ± 0.003 | 0.086 ± 0.004 |
| E only (-E-) | 0.957 ± 0.003 | 0.085 ± 0.004 |
| T only (–T) | 0.958 ± 0.003 | 0.084 ± 0.003 |
| M+E (ME-) | 0.961 ± 0.003 | 0.081 ± 0.003 |
| M+T (M-T) | 0.960 ± 0.003 | 0.082 ± 0.003 |
| E+T (-ET) | 0.960 ± 0.003 | 0.082 ± 0.003 |
| M+E+T (MET) | **0.963** ± 0.002 | **0.079** ± 0.003 |

**Table 19**. Augmentation-combination ablation on GridGraph (mean ± std over three seeds). "M" = node masking (10%), "E" = edge dropping (10%), "T" = topology perturbation (5%). Significant values are in [bold].

| Combination (M/E/T) | F1 | FNR |
|---|---|---|
| None (—) | 0.962 ± 0.003 | 0.044 ± 0.003 |
| M only (M–) | 0.969 ± 0.002 | 0.038 ± 0.003 |
| E only (-E-) | 0.971 ± 0.002 | 0.037 ± 0.002 |
| T only (–T) | 0.971 ± 0.002 | 0.037 ± 0.002 |
| M+E (ME-) | 0.972 ± 0.002 | 0.036 ± 0.002 |
| M+T (M-T) | 0.972 ± 0.002 | 0.036 ± 0.002 |
| E+T (-ET) | 0.971 ± 0.002 | 0.037 ± 0.002 |
| M+E+T (MET) | **0.973** ± 0.002 | **0.035** ± 0.002 |

**Table 20**. Augmentation-combination ablation on Texas2000 (mean ± std over three seeds). Component settings are the same as in Table 19. Significant values are in [bold].

| Layers | F1 | FNR | Epoch time (×) | Peak memory (×) |
|---|---|---|---|---|
| 1 | 0.956 ± 0.003 | 0.086 ± 0.004 | 0.78 | 0.90 |
| 2 | **0.963** ± 0.002 | **0.079** ± 0.003 | 1.00 | 1.00 |
| 3 | 0.961 ± 0.002 | 0.081 ± 0.003 | 1.28 | 1.15 |
| 4 | 0.958 ± 0.003 | 0.084 ± 0.004 | 1.55 | 1.30 |

**Table 21**. Depth study on GridGraph. The 2-layer setting corresponds to our main configuration. Significant values are in [bold].

| Layers | F1 | FNR | Epoch time (×) | Peak memory (×) |
|---|---|---|---|---|
| 1 | 0.966 ± 0.003 | 0.040 ± 0.003 | 0.79 | 0.91 |
| 2 | 0.973 ± 0.002 | 0.035 ± 0.002 | 1.00 | 1.00 |
| 3 | **0.974** ± 0.002 | **0.034** ± 0.002 | 1.28 | 1.16 |
| 4 | 0.971 ± 0.002 | 0.037 ± 0.002 | 1.56 | 1.31 |

**Table 22**. Depth study on Texas2000 under the same protocol. Significant values are in [bold].

| Dataset | Model | Acc. | F1 | Recall | FAR | FNR | Time (×GCN) | Mem. (×GCN) |
|---|---|---|---|---|---|---|---|---|
| GridGraph | GCFR–GCN (baseline) | 0.942 | 0.963 | 0.972 | 0.341 | 0.079 | 1.00 | 1.00 |
| | GCFR–GAT (1L, 4H) | **0.946** | **0.968** | **0.975** | **0.328** | **0.074** | ≈1.12 | ≈1.07 |
| Texas2000 | GCFR–GCN (baseline) | 0.974 | 0.973 | 0.967 | 0.023 | 0.035 | 1.00 | 1.00 |
| | GCFR–GAT (1L, 4H) | **0.976** | **0.977** | **0.975** | **0.022** | **0.031** | ≈1.14 | ≈1.09 |

**Table 23**. Encoder-only comparison under identical augmentation, loss, classifier, and schedule. "×GCN" denotes wall-time/memory overhead relative to the GCN run on the same machine. Significant values are in [bold].

the main setup, and we report the mean ± standard deviation over three seeds. The results are summarized in Tables 19 and 20.

Across both datasets, turning on any single augmentation improves over the "None" configuration, and pairwise combinations (M+E, M+T, E+T) recover most of the gains. The full combination (M+E+T) achieves the best F1 and the lowest FNR, but the margin over the best pair is small (about 0.2–0.3 percentage points on F1 and 0.001–0.002 on FNR), indicating limited but positive complementarity. On GridGraph, moving from no augmentation to M+E+T reduces FNR by 0.016 (0.095 → 0.079); On Texas2000, the reduction is 0.009 (0.044 → 0.035), consistent with the larger graph being more tolerant to moderate perturbations. The near-additive behavior aligns with the single-factor sensitivity trends reported earlier: Masking and edge dropping contribute most of the benefit, while topology perturbation adds a consistent but smaller improvement. For deployment, if augmentation must be simplified, M+E is a near-optimal two-component choice; Otherwise, keeping all three at their default strengths yields the best reliability profile with minimal additional complexity.

## Depth optimization of the GCN encoder

We investigate the effect of GCN depth on representation quality and efficiency. Under the unchanged pipeline (same augmentations, projection head, loss, batch size, and schedule), we vary the number of GCN layers $L \in \{1, 2, 3, 4\}$ while keeping the hidden dimension fixed. Results are reported as mean ± standard deviation over three seeds. We also present relative per-epoch wall-time and peak memory with respect to the 2-layer setting.

Tables 21, 22 reveal a consistent pattern. On GridGraph, the 1-layer model underfits (lower F1, higher FNR), the 2-layer configuration achieves the best overall balance, and deeper models (3–4 layers) do not improve F1 and slightly increase FNR, consistent with over-smoothing effects and noise propagation across neighborhoods. On the larger and more heterogeneous Texas2000 graph, 3 layers provide a marginal gain over 2 layers (F1 0.974 vs. 0.973; FNR 0.034 vs. 0.035), but 4 layers again degrade. Training cost scales nearly linearly with depth (about $+28\%$ and $+56\%$ time for 3 and 4 layers, respectively), with a similar trend in peak memory.

We keep 2 layers as the default for general deployments due to their strong reliability profile and efficiency. When compute budget permits and the graph is large with longer-range dependencies, 3 layers can be enabled to squeeze a small additional gain, while 1 and 4 layers are not recommended, given their underfitting and over-smoothing tendencies, respectively.

## Controlled evaluation of an attention-based encoder variant

We conduct a controlled, encoder-only study to assess the benefit of adaptive neighbor weighting without altering the GCFR pipeline. Specifically, we replace the 2-layer GCN aggregator with a 1-layer, 4-head multi-

head attention encoder (GCFR–GAT), while keeping the output dimension, augmentations (node masking, edge dropping, topology perturbation), InfoNCE loss, classifier head, and training schedule identical to the baseline. Results are reported as means over three random seeds with early stopping by validation F1. Table 23 summarizes performance and the relative per-epoch overhead.

**Overall performance.** Attention-based aggregation delivers modest yet consistent gains across both graphs. On Texas2000, F1/Recall improved from 0.973/0.967 to 0.977/0.975 with FNR reduced from 0.035 to 0.031. On GridGraph, F1/Recall rises from 0.963/0.972 to 0.968/0.975, while FNR drops from 0.079 to 0.074. These improvements align with the intuition that learned neighbor weighting can suppress noisy neighbors and emphasize links along plausible fault-propagation paths in heterogeneous neighborhoods.

**Computational overhead.** GCFR–GAT adds a constant-factor cost due to per-edge scoring and neighborhood softmax, increasing per-epoch wall-time by roughly 12–14% and peak memory by 7–9%, while the rest of the pipeline remains unchanged. This overhead is non-negligible for repeated retraining and large-graph inference.

**Rationale for retaining GCN as the default.** First, reliability at lower cost: the GCN-based GCFR already achieves a strong false-negative profile (low FNR), especially on Texas2000; GAT further lowers FNR only marginally, yielding diminishing returns under typical deployment budgets. Second, robustness–efficiency under homophily and label scarcity: many operational sub-networks exhibit moderate homophily and regular neighborhoods, where uniform aggregation is a strong inductive bias; in such regimes, the observed GAT gains (no more than 0.5–0.8 percentage points on F1/Recall in our runs) do not justify the extra overhead, and the variance-reducing, compute-efficient GCN is a safer default for periodic retraining and online scoring. Third, simplicity and scalability: the GCN encoder matches our near-linear training complexity in graph size and a lower memory footprint, simplifying scaling to larger PCNs and facilitating roll-outs on commodity GPUs. Accordingly, we retain GCN as the default encoder and position GAT as an optional plug-in for scenarios characterized by noisy or heterogeneous neighborhoods, where its modest gains justify the additional cost.

## Complexity comparison in time and space

We compare the complexity of GCFR with representative alternatives both analytically and empirically. Here $N$ and $M$ denote the numbers of nodes and edges of the sparse PCN graph, $L$ is the encoder depth, $D$ the input feature width, $F$ the hidden width, $C$ the number of classes, $p$ the projection dimension, and $b$ the mini-batch size for contrastive matching.

On sparse PCN graphs, a GCN encoder of depth $L$ costs $\mathcal{O}\big(L\,(MF + NF^2)\big)$ time per pass. GCFR encodes two augmented views, so the encoder term is roughly doubled; the projection head adds a low-order $\mathcal{O}(NFp)$ term. With mini-batch contrastive learning and negative sampling, the similarity computation among projected embeddings scales as $\mathcal{O}(b^2)$ (instead of $\mathcal{O}(N^2)$). A downstream classifier contributes $\mathcal{O}(NFC)$. Space-wise, the graph structure is stored once at $\mathcal{O}(N + M)$; model parameters are $\mathcal{O}(DF + F^2)$; the two per-node embeddings contribute $\mathcal{O}(2NF)$; per-layer node activations scale as $\mathcal{O}(LNF)$. Aggregating these yields an overall space complexity of $\mathcal{O}(N + M + NF)$, which simplifies to $\mathcal{O}(N + M)$ when $F \ll N$. For attention-style encoders on sparse graphs, the asymptotic order remains the same as GCN, while constant factors increase due to per-edge scoring and neighborhood softmax.

We report per-epoch wall time and peak memory relative to the GCFR default configuration (two-layer GCN with InfoNCE). All runs use identical hardware, data loaders, and training schedules; values are means over three seeds. Table 24 summarizes training cost and memory footprint across variants, and Table 25 reports inference-time overhead for uncertainty modules.

| Method / Variant | Time (×) | Peak memory (×) | Notes |
|---|---|---|---|
| GCFR (2-layer GCN, InfoNCE) | 1.00 | 1.00 | Default |
| + Temporal consistency (TempReg) | 1.05 | 1.01 | Snapshot consistency plug-in |
| + Domain alarm features (+Feat) | 1.02 | 1.00 | Small attribute concat |
| + Feat + DWT denoising | 1.04 | 1.01 | Wavelet preprocessor |
| SimCLR-style objective | 1.01 | 1.00 | Same views/projection |
| BYOL-style objective | 1.13 | 1.07 | Momentum target + predictor |
| GAT encoder (1-layer) | 1.12 | 1.07 | Attention mixing |
| GCN depth $L$=3 | 1.28 | 1.16 | Deeper receptive field |
| GCN depth $L$=4 | 1.56 | 1.31 | Over-smoothing regime |
| DWT+LSTM baseline | 0.92 | 0.96 | Sequence model on denoised alarms |
| DWT+ANN baseline | 0.85 | 0.92 | MLP on wavelet energy |
| NSMFD baseline | 0.88 | 0.90 | Classical feature/classifier pipeline |
| Temporal attention encoder (TGAT-like) | 1.18 | 1.11 | Time-aware mixing |

**Table 24**. Training cost and memory footprint (relative multipliers).

| Method | Time (×) | Peak memory (×) | Notes |
|---|---|---|---|
| Temperature scaling | 1.02 | 1.00 | post-hoc calibrator |
| MC dropout ($T=10$ samples) | 6.5 | 1.05 | Bayesian approximation |
| Deep ensemble ($M=3$ models) | 3.1 | 1.10 | Bayesian model averaging |

**Table 25**. Inference-time overhead for uncertainty modules (relative to a single deterministic model).

For decision-time add-ons, the training pipeline remains unchanged; we summarize inference overhead below in Table 25.

As seen in Table 24, GCFR shares the same asymptotic order as standard GCN-based methods; its constant factors come from processing two augmented views and the projection head, while the in-batch similarity term is minor on sparse PCNs. Among variants, SimCLR-style is nearly cost-neutral; temporal consistency and DWT-based denoising add small constant factors; BYOL-style and attention encoders increase cost due to additional computation (momentum target/predictor or attention mixing); deeper GCNs scale almost linearly and may suffer from over-smoothing. At inference (Table 25), temperature scaling provides near-free calibration, while small Bayesian approximations (ensembles or MC dropout) trade compute for improved calibration and selective prediction.

## Conclusion

This work presented GCFR, a deployment-oriented framework for fault diagnosis in power communication networks that learns robust node representations via multi-view graph contrast and a lightweight classifier. Beyond the base design, we incorporated a temporal-consistency regularizer to stabilize embeddings under topology evolution, domain-aware alarm-propagation features, and a lightweight DWT denoiser to withstand timestamp jitter, missing events, and spurious cascades, as well as approximate uncertainty calibration with selective prediction to explicitly control the coverage–risk trade-off. Comprehensive experiments on two benchmark graphs and an anonymized real-world slice show that GCFR maintains strong recall under scarce labels, improves robustness to topology churn and alarm noise, and consistently outperforms strong supervised and signal-processing baselines with modest training and inference overhead.

We focus on practical aspects that facilitate broader adoption. Future work includes expanding case studies to additional regions and seasons, enriching visualization and interpretability tools for operator training, packaging GCFR with preset configurations and monitoring dashboards, and conducting systematic latency/energy profiling across diverse hardware backends. We also plan to broaden comparisons with recent graph encoders and self-supervised objectives, and to run user studies on workflow integration and alert-triage ergonomics.

## Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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## Author contributions

B.C. and G.L. conceived the core idea of the GCFR framework and designed the methodology. B.C. wrote the main manuscript text and performed the experiments. Y.Z. and C.L. contributed to the implementation of the data augmentation strategies and the construction of the benchmark datasets. B.Q. was responsible for the result analysis and performance evaluation. J.L. supervised the project and provided critical revisions to the manuscript. All authors reviewed and approved the final version of the manuscript.

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## Competing interests

The authors declare no competing interests.

## Additional information

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